SAT Math Mastery · Session 3

Nonlinear Equations and Systems

Time	3 hours: Part A in class (about 2 hours of solving), then Part B, the self-assessment, done independently (about 50 minutes), and a short error log.
Topics	solving quadratics: factoring, completing the square, discriminant, sum and product of roots · line and parabola, line and circle systems · number of solutions and unknown constants · rational equations and extraneous solutions · exponential equations · absolute value, radical and higher-degree equations
Name / Date	_____ _____

PART A · IN CLASS — 35 PROBLEMS

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal). Grey boxes show exactly what to type in Desmos, one expression per line; tap a grey dot to read an intersection, intercept or vertex. A letter such as a, c or k becomes a slider when you accept Desmos's offer to add one.

1. Medium · Desmos

$$\frac{1}{x^2+10x+25}=4$$

Which of the following could be the value of $x+5$?

(A) $\frac{1}{2}$ ✓ (B) $\frac{5}{2}$

(C) $\frac{9}{2}$ (D) $\frac{11}{2}$

1. Type `y=1/(x^2+10x+25)` and `y=4`.

2. Tap the two intersections: $(-5.5, 4)$ and $(-4.5, 4)$. These are the values of x.

3. The question asks for $x+5$. Type `-5.5+5`: it returns -0.5 and type `-4.5+5`: it returns 0.5.

4. Only $\frac{1}{2}$ is among the options, so the answer is (A).

Answer: (A)

2. Hard · Desmos

In the xy-plane, the graphs of $y=x^2-6x+8$ and $y=2x-4$ intersect at two points. What is the sum of the x-coordinates of these two points?

1. Type `y=x^2-6x+8` and `y=2x-4`.

2. Tap the two intersections: (2, 0) and (6, 8).

3. Type `2+6`: it returns 8, the sum of the x-coordinates.

Answer: 8

3. HARD · DESMOS

In the xy-plane, the graphs of $y = x^2 - 4x - 5$ and $y = 3x + 1$ intersect at two points. What is the product of the x-coordinates of these two points?

1. Type `y=x^2-4x-5` and `y=3x+1`.
2. Tap the two intersections: $(-0.772, -1.316)$ and $(7.772, 24.316)$.
3. The x-values are not whole numbers, so use the exact forms $\frac{7 \pm \sqrt{73}}{2}$. Type `(7+sqrt(73))/2*(7-sqrt(73))/2`: it returns -6.
4. That is the product of the x-coordinates: -6.

Answer: -6

4. HARD · DESMOS

In the xy-plane, the graphs of $y = x^2 - x - 14$ and $y = 2x - 4$ intersect at two points. What is the positive difference between the x-coordinates of these two points?

1. Type `y=x^2-x-14` and `y=2x-4`.
2. Tap the two intersections: $(-2, -8)$ and $(5, 6)$.
3. Type `5-(-2)`: it returns 7, the positive difference of the x-coordinates.

Answer: 7

5. HARD · DESMOS

Suppose m and n are positive integers such that $mn > 20$ and $12mn$ is a perfect square. What is the least possible value of mn?

1. $12mn$ is a perfect square exactly when $mn = 3k^2$. Type `3*[1...5]^2`: Desmos lists the possible values of mn: 3, 12, 27, 48, 75.
2. The first value above 20 is 27.
3. Type `sqrt(12*27)`: it returns 18, a whole number, so $12mn$ is a perfect square.
4. Compare a value that is not on the list: type `sqrt(12*24)`: it returns 16.9706, not a whole number. So the least value is 27.

Answer: 27

6. HARD · DESMOS

If $x^2 - 8x + 5 = 0$ has roots r and s, what is the value of $r^2 + s^2$?

1. Type `y=x^2-8x+5` and tap its two zeros: (0.6834, 0) and (7.3166, 0) (these are r and s).
2. They are $4 \pm \sqrt{11}$. Type the exact squares: Type `(4-sqrt(11))^2+(4+sqrt(11))^2`: it returns 54
3. Check with the formula $(r + s)^2 - 2rs$: Type `8^2-2*5`: it returns 54. So $r^2 + s^2 = 54$.

Answer: 54

7. HARD · DESMOS

What is the positive solution to the equation $x^2 - 6x - 7 = 0$, found by completing the square?

1. Completing the square turns $x^2 - 6x - 7 = 0$ into $(x-3)^2 = 16$. Type `y=(x-3)^2` and `y=16`.
2. Tap the two intersections: $(-1, 16)$ and $(7, 16)$.
3. The positive solution is $x = 7$. Check: Type `7^2-6*7-7`: it returns 0.

Answer: 7

8. HARD · DESMOS

If $y = x^2 - 7$ and $x + y = 5$, what is the value of x^2y^2 for the solution (x, y) of this system with $x > 0$?

1. Rewrite $x + y = 5$ as $y = 5 - x$. Type `y=x^2-7` and `y=5-x`.
2. Tap the two intersections: $(-4, 9)$ and $(3, 2)$. We need $x > 0$, so $(x, y) = (3, 2)$.
3. Type `3^2*2^2`: it returns 36. So $x^2y^2 = 36$.

Answer: 36

9. HARD · DESMOS

A taxi company uses the formula $10p = q + 25r$, where p is the fare per passenger, in dollars; q is a fixed service charge, in dollars; and r is the distance traveled, in kilometers. Which equation correctly expresses p in terms of q and r?

(A) $p = \dfrac{q}{10} + 25r$

(B) $p = q + \dfrac{25r}{10}$

(C) $p = 10\,(q + 25r)$

(D) $p = \dfrac{q + 25r}{10}$ ✓

1. Test numbers: let $q = 20$ and $r = 4$. Type `20+25*4`: it returns 120, so $10p = 120$ and $p = 12$ (type `(20+25*4)/10`: it returns 12).
2. Test option $\dfrac{q + 25r}{10}$: Type `(20+25*4)/10`: it returns 12, a match.
3. Test the others: Type `20/10+25*4`: it returns 102, type `20+25*4/10`: it returns 30 and type `10*(20+25*4)`: it returns 1200, none equal to 12.
4. Only $p = \dfrac{q + 25r}{10}$ works, which is option (D).

Answer: (D)

10. HARD · DESMOS

A company's monthly profit, in thousands of dollars, is modeled by $P(x) = -2x^2 + 40x - 150$, where x is the price of the product in dollars. For what price x, other than $x = 5$, does the company break even (that is, $P(x) = 0$)?

1. Break even means $P(x) = 0$. Type `y=-2x^2+40x-150` (letting x stand for the price).

2. Tap the two zeros: (5, 0) and (15, 0).

3. Other than $x = 5$ the price is 15. Check: Type `-2*15^2+40*15-150`: it returns 0.

Answer: 15

11. HARD · DESMOS

A system of equations is given below, where x represents the number of years since a company's founding and y represents its annual revenue, in millions of dollars.

$$y = x^2 - 5x + 20$$

$$y = 3x + 8$$

For how many values of x, with $x \geq 0$, are the two models equal?

(A) 0 (B) 1

(C) 2 ✓ (D) 3

1. Type `y=x^2-5x+20` and `y=3x+8`.

2. Tap the two intersections: (2, 14) and (6, 26).

3. Both have $x \geq 0$, so the models are equal for 2 values of x. The answer is (C).

Answer: (C)

12. HARD · DESMOS

$$x^2\,(x + 5)\,(x - b) = 0$$

In the given equation, b is a positive constant. The sum of the solutions of the equation is 7. What is the value of b?

1. Type `y=x^2*(x+5)*(x-b)` (accept the slider for b).

2. Slide b and tap the zeros on the x-axis. At $b = 12$ they are $(-5, 0)$, $(0, 0)$ and $(12, 0)$ (the zero at 0 is counted once).

3. Type `-5+0+12`: it returns 7, which is the required sum 7. So $b = 12$.

Answer: 12

13. HARD · DESMOS

If the graphs of $y = x^2 + 1$ and $y = -5x + p$ intersect at the point $(2, r)$, what is the value of $\frac{p}{r}$?

1. Type `y=x^2+1` and tap the point on it with $x = 2$: (2, 5). So $r = 5$.

2. Type `y=-5x+p` (accept the slider for p) and slide p until the line passes through (2, 5): $p = 15$. (At $p = 15$ the line is at height type `-5*2+15`: it returns 5 when $x = 2$.)

3. Type `15/5`: it returns 3.

Answer: 3

14. HARD · DESMOS

The functions f and h are defined by $f(x) = x^2 + 3x - 40$ and $h(x) = x^2$. If $3h(x) = f(2x)$, what is the positive value of x?

1. $f(2x)$ means putting $2x$ in for x in f. Type `y=3x^2` ($3h(x)$) and `y=(2x)^2+3(2x)-40` ($f(2x)$).

2. Tap the two intersections: (−10, 300) and (4, 48).

3. The positive value is $x = 4$. Check: Type `3*4^2`: it returns 48 and type `(2*4)^2+3*(2*4)-40`: it returns 48.

Answer: 4

15. HARD · DESMOS

Let $f(x) = ax^3$ and $g(x) = bx^5$ for any value of x. If a and b are positive constants, for how many values of x is $f(x) = g(x)$?

(A) None

(B) One

(C) Two

(D) Three ✓

1. Type `y=a*x^3` and `y=b*x^5` (accept sliders for a and b, and keep both positive).

2. With $a = 1$ and $b = 1$ tap the intersections: (−1, −1), (0, 0) and (1, 1).

3. Change the sliders to $a = 2$ and $b = 0.5$: the intersections are (−2, −16), (0, 0) and (2, 16). Still three.

4. For any positive a and b there are 3 values of x (the answer is (D)).

Answer: (D)

16. HARD · DESMOS

If $(x + 3)^2 = 1$ and $(x - 4)^2 = 36$, what is the value of x?

1. Type `y=(x+3)^2` and `y=1`. Tap the intersections: (−4, 1) and (−2, 1).

2. Type `y=(x-4)^2` and `y=36`. Tap the intersections: (−2, 36) and (10, 36).

3. The only x in both lists is −2. Check: Type `(-2+3)^2`: it returns 1 and type `(-2-4)^2`: it returns 36.

Answer: −2

17. HARD · DESMOS

What value of x satisfies the equation below?

$$\frac{x}{x-3}+\frac{2}{x}=\frac{3}{x-3}$$

1. Type `y=x/(x-3)+2/x` and `y=3/(x-3)`.
2. Tap the intersection: $(-2, -0.6)$. (There is no point at $x=3$: both graphs have a break there.)
3. Check $x=-2$: Type `-2/(-2-3)+2/(-2)`: it returns -0.6 and type `3/(-2-3)`: it returns -0.6.

Answer: -2

18. HARD · DESMOS

What value of x satisfies the equation $2^{3x}\cdot 4^{x+1}=8^{x+3}$?

1. Write both sides with base 2: the exponents are $3x+2(x+1)$ on the left and $3(x+3)$ on the right. Type `y=3x+2(x+1)` and `y=3(x+3)`.
2. Tap the intersection: (3.5, 19.5). Its x-value is the solution, $x=3.5$.
3. Check in the original equation: Type `2^(3*3.5)*4^(3.5+1)`: it returns 741455.2002 and type `8^(3.5+3)`: it returns 741455.2002. The answer is $\frac{7}{2}$.

Answer: $\frac{7}{2}$ **(or** 3.5**)**

19. HARD · DESMOS

$$y=x^2-4x+1$$
$$y=2x-8$$

How many solutions does the system above have?

(A) Zero

(B) Exactly one ✓

(C) Exactly two

(D) Infinitely many

1. Type `y=x^2-4x+1` and `y=2x-8`.
2. The line just touches the parabola at one point: $(3, -2)$.
3. Confirm with the discriminant of $x^2-6x+9=0$: Type `(-6)^2-4*1*9`: it returns 0, so there is exactly one solution, option (B).

Answer: (B)

20. HARD · DESMOS

The equation $2x^2 - 12x + k = 0$, where k is a constant, has two real solutions. One solution is twice the other solution. What is the value of k?

1. Type `y=2x^2-12x+k` (accept the slider for k).

2. Slide k until the two zeros are in the ratio $2:1$. At $k = 16$ they are $(2, 0)$ and $(4, 0)$.

3. Type `4/2`: it returns 2, so one solution is twice the other. Hence $k = 16$.

Answer: 16

21. HARD · DESMOS

For a real number x, $4^x - 2^{x+2} - 32 = 0$. What is the value of 2^{x+1}?

(A) -8 (B) 8

(C) 16 ✓ (D) 32

1. Type `y=4^x-2^(x+2)-32`.

2. Tap the zero: $(3, 0)$. So $x = 3$.

3. The question asks for 2^{x+1}: Type `2^(3+1)`: it returns 16. The answer is (C).

Answer: (C)

22. HARD · DESMOS

The equation $|x + 6| - 7 = \frac{x}{3} + c$, where c is a constant, has exactly one solution x. What is the value of $2c + x$?

(A) -11 (B) -6

(C) -5 **(D) -16 ✓**

1. Type `y=abs(x+6)-7` and `y=x/3+c` (accept the slider for c). Tap the corner of the V: $(-6, -7)$.

2. Slide c. The line is gentler than both arms of the V: with c above the right value it cuts the V twice, with c below it misses the V. It cuts exactly once only when it passes through the corner.

3. At the corner $x = -6$ the line has height $-6/3 + c$, which must equal -7. Type `-7-(-6/3)`: it returns -5, so $c = -5$ (the slider shows the line through $(-6, -7)$).

4. The one solution is $x = -6$. Type `2*(-5)+(-6)`: it returns -16. The answer is (D).

Answer: (D)

23. HARD · DESMOS

In the xy-plane, the line $y = mx - 5$ intersects the parabola $y = x^2 - 6x + 11$ at exactly one point, where m is a negative constant. What is the value of m?

(A) 14 **(B) -14 ✓**

(C) -2 (D) 2

1. Type `y=x^2-6x+11` and `y=m*x-5` (accept the slider for m).
2. Slide m until the line just touches the parabola. This happens at $m = 2$, touching at (4, 3), and at $m = -14$, touching at $(-4, 51)$.
3. The problem says m is negative, so $m = -14$. Check the touch point: Type `(-4)^2-6*(-4)+11`: it returns 51 and type `-14*(-4)-5`: it returns 51. The answer is (B).

Answer: (B)

24. HARD · DESMOS

$$x^2 + y^2 = 65$$
$$x - y = 7$$

The solution (a, b) to the system of equations above satisfies $a > 0$. What is the value of $a + 3b$?

1. Rewrite the second equation as $y = x - 7$. Type `x^2+y^2=65` (a circle) and `y=x-7`.
2. Tap the two points where the line crosses the circle: $(-1, -8)$ and (8, 1). The problem wants $a > 0$.
3. So $(a, b) = (8, 1)$. Type `8+3*1`: it returns 11.

Answer: 11

25. HARD · DESMOS

In the xy-plane, the graphs of $y = x^2 + 6x + 4$ and $y = mx - 4$ have no point in common, where m is a constant. What is the sum of all integer values of m?

1. Set the expressions equal: $x^2 + 6x + 4 = mx - 4$, so $x^2 + (6 - m)x + 8 = 0$. No common point means no real solution, so the discriminant must be negative: $(6 - m)^2 - 32 < 0$.
2. Type `y=(6-x)^2-32` (letting x stand for m).
3. Tap the two zeros: (0.3431, 0) and (11.6569, 0). The curve is below the axis (negative discriminant) between them, so $0.3431 < m < 11.6569$.
4. The integers in that range are $1, 2, \ldots, 11$. Add them: Type `11*12/2`: it returns 66.

Answer: 66

26. HARD · DESMOS

$$2^x = 3x + k$$

In the equation above, k is a constant. What is the smallest integer value of k for which the equation has two distinct real solutions?

(A) -2 ✓ (B) -1

(C) 0 (D) -3

1. Rewrite the equation as $2^x - 3x = k$. Type `y=2^x-3x` and tap its lowest point: (2.1137, −2.0131).
2. Each value of k is a horizontal line. Type `y=-3`: it is below the lowest point, so it never meets the curve (no solutions).
3. Type `y=-2`: it is just above the lowest point and meets the curve twice, at (2, −2) and (2.2245, −2).
4. Two solutions need $k > -2.0131$, so the smallest integer is -2. The answer is (A).

Answer: (A)

27. HARD · DESMOS

For an integer k, the equation

$$9 - 3\,|2x - 8| = 1 - \frac{5k}{2}$$

has no real solution x. What is the greatest possible value of k?

1. Type `y=9-3abs(2x-8)` and tap its highest point: (4, 9).
2. Type `y=1-5k/2` (accept the slider for k). At $k = -3$ the line is at height 8.5 and meets the graph at (3.9167, 8.5) and (4.0833, 8.5).
3. At $k = -4$ the line is higher than the top of the graph: Type `1-5*(-4)/2`: it returns 11, which is above 9, so there is no solution.
4. So the greatest integer with no solution is -4.

Answer: -4

28. HARD · DESMOS

For a constant k, the equation

$$2\sqrt{x+1} + 2\sqrt{15 - x} = k$$

has exactly one real solution. What is the value of k?

(A) 16 (B) $2 + 2\sqrt{15}$

(C) 8 **(D) $8\sqrt{2}$ ✓**

1. Type `y=2*sqrt(x+1)+2*sqrt(15-x)`. It only exists for $-1 \le x \le 15$. Tap the left end (−1, 8), the top (7, 11.3137) and the right end (15, 8).
2. The solutions are where the horizontal line $y = k$ meets the curve. Type `y=11`: it meets the curve twice, at (3.3621, 11) and (10.6379, 11).
3. Type `y=8`: it meets the curve at both ends, again twice. A line meets the curve exactly once only when it just touches the top.
4. Type `4*sqrt(8)`: it returns 11.3137, which is the top height. Type `8*sqrt(2)`: it returns 11.3137, the same number, so $k = 8\sqrt{2}$. The answer is (D).

Answer: (D)

29. HARD · DESMOS

$$x^2 + y^2 = 45$$
$$y = mx + 15$$

In the system above, m is a negative constant. If the system has exactly one solution (x, y), what is the value of x?

(A) -3 (B) 3

(C) 6 ✓ (D) -6

1. Type `x^2+y^2=45` (a circle) and `y=m*x+15` (accept the slider for m).
2. Slide m (negative) until the line touches the circle at exactly one point. At $m = -2$ the point is (6, 3).
3. Check: Type `6^2+3^2`: it returns 45 and type `-2*6+15`: it returns 3. So $x = 6$, option (C).
Answer: (C)

30. HARD · DESMOS

$$y = 3x^2 + 4$$
$$y = 1 - kx$$

In the system above, k is a constant. For which of the following values of k does the system have two distinct solutions (x, y) whose x-coordinates are both negative?

(A) 6 **(B) 7 ✓**

(C) -7 (D) 2

1. Type `y=3x^2+4` and `y=1-k*x` (accept the slider for k).
2. Set $k = 7$: the line meets the parabola at $(-1.7676, 13.3731)$ and $(-0.5657, 4.9602)$, both with negative x.
3. Try the other options: $k = 6$ only touches, at $(-1, 7)$; $k = 2$ never meets the parabola; $k = -7$ meets it at two points with positive x.
4. So the answer is $k = 7$, option (B).
Answer: (B)

31. HARD · DESMOS

For how many integer values of k with $-10 \le k \le 10$ does the equation

$$\sqrt{x+k} = x - 2$$

have exactly one real solution?

1. Square both sides: $x + k = (x-2)^2$, and remember $x - 2 \ge 0$, because a square root is never negative. Solve for k: $k = x^2 - 5x + 4$ with $x \ge 2$.
2. Type `y=x^2-5x+4{x>=2}`. Tap the left end $(2, -2)$ and the lowest point $(2.5, -2.25)$. The number of solutions is the number of times the horizontal line $y = k$ meets this curve.
3. Type `y=-2`: it meets the curve twice, at $(2, -2)$ and $(3, -2)$, so $k = -2$ gives two solutions. Type `y=-1`: it meets the curve only once, at $(3.618, -1)$.
4. Every line from $y = -1$ upward meets the curve exactly once. The values are $k = -1, 0, 1, \ldots, 10$. Count them: Type `10-(-1)+1`: it returns 12. (The single touch at $k = -2.25$ is not an integer.)

Answer: 12

32. HARD · DESMOS

In the xy-plane, the line $y = kx+4$, where k is a constant, is tangent to the parabola $y = x^2+4x+13$ for exactly two values of k. For each of these values the line touches the parabola at one point. What is the greatest y-coordinate of these two points of tangency?

(A) 10 (B) 13

(C) 34 ✓ (D) 3

1. Type `y=x^2+4x+13` and `y=k*x+4` (accept the slider for k).
2. Slide k until the line just touches the parabola. This happens at $k = -2$, touching at $(-3, 10)$, and at $k = 10$, touching at $(3, 34)$.
3. The greater y-coordinate is 34 (option (C)). Check: Type `3^2+4*3+13`: it returns 34.

Answer: (C)

33. HARD · DESMOS

For a real number k, consider the equation

$$x - 4\sqrt{x} = k.$$

For how many integer values of k from -10 to 10, inclusive, does the equation have exactly one real solution x?

1. Type `y=x-4*sqrt(x)`. Tap the left end $(0, 0)$, the lowest point $(4, -4)$ and the right-hand zero $(16, 0)$. The solutions of the equation are where the horizontal line $y = k$ meets this curve.
2. Type `y=-4`: it only touches the lowest point, so exactly one solution. Type `y=-2`: it meets the curve twice, at $(0.3431, -2)$ and $(11.6569, -2)$. Type `y=0`: twice again, at $(0, 0)$ and $(16, 0)$. Below $y = -4$ there are no solutions.
3. Type `y=3`: only once, at $(21.583, 3)$. Every line above $y = 0$ meets the curve only once, on the rising part to the right of $x = 16$.
4. Exactly one solution happens for $k = -4$ (one value) and for $k = 1, 2, \ldots, 10$ (ten values). Count them: Type `1+10`: it returns 11.

Answer: 11

34. HARD · DESMOS

$$x^3 + (3-k)x^2 - (10+3k)x + 10k = 0$$

In the equation above, k is a constant. The equation has exactly two distinct real solutions for exactly two values of k. What is the sum of these two values of k?

1. Type `y=x^3+(3-k)x^2-(10+3k)x+10k` (accept the slider for k). The solutions are the zeros of the graph.

2. At $k = 0$ there are three zeros: $(-5, 0)$, $(0, 0)$ and $(2, 0)$.

3. At $k = 2$ and at $k = -5$ the graph has only two zeros, $(-5, 0)$ and $(2, 0)$, because one of them is a touching zero.

4. Type `2+(-5)` : it returns -3, the sum of the two values of k.

Answer: -3

35. HARD · DESMOS

$$|2x - k| + k = x + 6$$

In the equation above, k is a constant. The equation has exactly one solution. What is the value of k?

1. Type `y=abs(2x-k)+k` and `y=x+6` (accept the slider for k). The first graph is a V whose corner is at $\left(\frac{k}{2}, k\right)$, and the line $y = x + 6$ is gentler than both arms.

2. Set $k = 0$: the line cuts the V twice, at $(-2, 4)$ and $(6, 12)$. Slide k up and the two points move together.

3. At $k = 12$ the corner $(6, 12)$ sits on the line and there is exactly one intersection. For k larger than that the V is above the line and they never meet.

4. Check with the corner: $\left(\frac{k}{2}, k\right)$ lies on $y = x + 6$ when $k = \frac{k}{2} + 6$. Type `12/2+6` : it returns 12, which matches $k = 12$.

Answer: 12

PART B · SELF-ASSESSMENT — 22 PROBLEMS

36. MEDIUM · DESMOS

Emma is 4 years older than Grace. The square of Emma's age, in years, is 64 greater than the square of Grace's age. What is the sum, in years, of Emma's and Grace's ages?

1. Let x be Grace's age, so Emma's age is $x+4$. Type `y=(x+4)^2` (the square of Emma's age) and `y=x^2+64` (the square of Grace's age plus 64).
2. Tap the intersection: (6, 100). Its x-value is Grace's age.
3. Emma is $6+4$: Type `6+4`: it returns 10.
4. Type `6+10`: it returns 16. The sum of their ages is 16.

Answer: 16

37. MEDIUM · DESMOS

As part of a tourism marketing campaign, the number of visitors (in thousands) x weeks after the campaign begins is modeled by $v = x^2 + 2x - 3$. A separate baseline estimate predicts $v = -3$ for the same period. For how many nonnegative values of x are the two estimates equal?

(A) 0

(B) 1 ✓

(C) 2

(D) 3

1. Type `y=x^2+2x-3` and `y=-3`.
2. Tap the two intersections: $(-2, -3)$ and $(0, -3)$.
3. Only $x = 0$ is nonnegative, so the estimates are equal for 1 value of x: option (B).

Answer: (B)

38. MEDIUM · DESMOS

What is the positive solution to the equation $x^2 - 5x - 24 = 0$?

1. Type `y=x^2-5x-24`.
2. Tap the two zeros: $(-3, 0)$ and $(8, 0)$.
3. The positive solution is 8. Check: Type `8^2-5*8-24`: it returns 0.

Answer: 8

39. MEDIUM · DESMOS

What are the solutions to the equation $2x^2 - 5x - 3 = 0$?

(A) $x = -\frac{1}{2}$ and $x = 3$ ✓

(B) $x = \frac{1}{2}$ and $x = -3$

(C) $x = -\frac{3}{2}$ and $x = 1$

(D) $x = \frac{3}{2}$ and $x = -1$

1. Type `y=2x^2-5x-3`.
2. Tap the two zeros: $(-0.5, 0)$ and $(3, 0)$.
3. The solutions are $x = -\frac{1}{2}$ and $x = 3$, option (A). Check: Type `2*3^2-5*3-3`: it returns 0.

Answer: (A)

40. Medium · Desmos

If x and y are positive integers such that $x^2 - y^2 = 45$ and $x - y = 5$, what is the value of $x + y$?

(A) 7

(B) 9 ✓

(C) 11

(D) 13

1. Type `x^2-y^2=45` and `y=x-5` (from $x - y = 5$).

2. Tap the intersection: (7, 2). Both coordinates are positive integers.

3. Type `7+2`: it returns 9. So $x + y = 9$, option (B).

Answer: (B)

41. Medium · Desmos

An integer n is subtracted from its square. Which of the following could NOT be the result?

(A) 0

(B) An even integer

(C) An odd integer ✓

(D) A positive integer

1. Type `[-3...4]^2-[-3...4]`: it lists $n^2 - n$ for $n = -3, \ldots, 4$: 12, 6, 2, 0, 0, 2, 6, 12.

2. Look at the remainders after dividing by 2: type `mod([-3...4]^2-[-3...4],2)`: it returns 0, 0, 0, 0, 0, 0, 0, 0.

3. Every remainder is 0, so the result is always even. It can be 0, an even integer or a positive integer, but never odd: option (C).

Answer: (C)

42. Medium · Desmos

For which value of k does the equation $x^2 + 6x + k = 0$ have exactly one real solution?

(A) 3

(B) 6

(C) 9 ✓

(D) 12

1. Type `y=x^2+6x+k` (accept the slider for k).

2. Slide k. For $k = 3$ the parabola crosses the x-axis twice, but at $k = 9$ it only touches it at $(-3, 0)$.

3. Confirm with the discriminant: Type `6^2-4*1*9`: it returns 0. So $k = 9$, option (C).

Answer: (C)

43. Medium · Desmos

What is the solution set of the equation $\dfrac{x}{x-2} = \dfrac{2}{x-2} + 3$?

(A) $x = 2$ only

(B) $x = 4$ only

(C) $x = 2$ and $x = 4$

(D) There is no solution. ✓

1. Type `y=x/(x-2)` and `y=2/(x-2)+3`.
2. The graphs never meet: no intersection dot appears. Check the gap at a few x-values: Type `4/(4-2)-(2/(4-2)+3)`: it returns -2 and type `10/(10-2)-(2/(10-2)+3)`: it returns -2.
3. The difference is always -2, never 0, so there is no solution: option (D).

Answer: (D)

44. Medium · Desmos

Which inequality is equivalent to $8x - 12y > 20$?

(A) $2x - 3y > 5$ ✓

(B) $3x - 2y > 5$

(C) $4x - 3y > 5$

(D) $2y - 3x > 5$

1. Type `8x-12y>20`. Desmos shades the region.
2. Type `2x-3y>5`: the shading is exactly the same, so this is equivalent.
3. Type `3x-2y>5`: the shading differs. Test the point $(2, 0)$: Type `3*2-2*0`: it returns 6 (more than 5) but type `8*2-12*0`: it returns 16 (not more than 20).
4. So the equivalent inequality is $2x - 3y > 5$, option (A).

Answer: (A)

45. Medium · Desmos

How many distinct real solutions does the equation $4x^2 - 20x + 25 = 0$ have?

(A) Exactly one ✓

(B) Exactly two

(C) Infinitely many

(D) Zero

1. Type `y=4x^2-20x+25`.
2. The graph touches the x-axis at one point only: (2.5, 0).
3. Confirm with the discriminant: Type `(-20)^2-4*4*25`: it returns 0. Exactly one real solution: option (A).

Answer: (A)

46. Medium · Desmos

In the system of equations below, x and y satisfy both equations.

$$y = 5x + 2$$

$$y = 5x^2 + 2$$

Which ordered pair (x, y) is a solution to this system?

(A) $(0, 0)$

(B) $(0, 5)$

(C) $(1, 5)$

(D) $(1, 7)$ ✓

1. Type `y=5x+2` and `y=5x^2+2`.
2. Tap the intersections: (0, 2) and (1, 7).
3. Only $(1, 7)$ is among the options, so the answer is (D). (The point $(0, 5)$ fails: Type `5*0+2`: it returns 2, not 5.)
Answer: (D)

47. Medium · Desmos

$$7|x| = 63$$

What is the positive solution to the given equation?

1. Type `y=7abs(x)` and `y=63`.
2. Tap the intersections: (−9, 63) and (9, 63).
3. The positive solution is $x = 9$.
Answer: 9

48. Medium · Desmos

What is the sum of the y-coordinates of the solutions to the system below?

$$y = x^2 - 4x$$

$$y = 2x - 5$$

(A) -2

(B) 0

(C) 2 ✓

(D) 4

1. Type `y=x^2-4x` and `y=2x-5`.
2. Tap the intersections: (1, −3) and (5, 5).
3. Type `-3+5`: it returns 2, the sum of the y-coordinates: option (C).
Answer: (C)

49. Medium · Desmos

The height of a triangle is 4 cm more than its base. If the area of the triangle is 30 cm^2, what is the length of the base, in cm?

(A) 4

(B) 5

(C) 6 ✓

(D) 8

1. Base x and height $x+4$ give area $\frac{x(x+4)}{2}$. Type `y=x*(x+4)/2` and `y=30`.

2. Tap the intersections: $(-10, 30)$ and $(6, 30)$. A length is positive.

3. The base is 6 cm, option (C). Check: Type `6*(6+4)/2`: it returns 30.

Answer: (C)

50. Medium · Desmos

What is the solution to the equation $\frac{x+5}{x-1} = 3$, for $x \neq 1$?

1. Type `y=(x+5)/(x-1)` and `y=3`.

2. Tap the intersection: (4, 3).

3. Check: Type `(4+5)/(4-1)`: it returns 3. So $x = 4$.

Answer: 4

51. Medium · Desmos

The curve $y = \frac{x^2}{3}$ and the line $y = \frac{2x}{3}$ intersect at the origin and at the point (a, b). What is the value of b?

1. Type `y=x^2/3` and `y=2x/3`.

2. Tap the intersections: (0, 0) (the origin) and (2, 1.3333).

3. So $a = 2$ and b is the height: Type `2*2/3`: it returns 1.3333, which is $\frac{4}{3}$.

Answer: $\frac{4}{3}$

52. Medium · Desmos

For what values of x is the expression $2x^2 - 2x - 24$ equal to 0?

(A) $x = -4,\ x = 3$

(B) $x = 4,\ x = -3$ ✓

(C) $x = 4,\ x = 3$

(D) $x = -4,\ x = -3$

1. Type `y=2x^2-2x-24`.

2. Tap the two zeros: $(-3, 0)$ and $(4, 0)$.

3. The solutions are $x = 4$ and $x = -3$, option (B). Check: Type `2*4^2-2*4-24`: it returns 0.

Answer: (B)

53. Medium · Desmos

If $(x-3)(x+7) = -25$, what is the value of x?

1. Type `y=(x-3)(x+7)` and `y=-25`.

2. The line touches the lowest point of the parabola: $(-2, -25)$. So $x = -2$.

3. Check: Type `(-2-3)*(-2+7)`: it returns -25.

Answer: -2

54. Medium · Desmos

The square of a positive number is 12 more than 4 times the number. What is the number?

1. Type `y=x^2` (the square) and `y=4x+12` (4 times the number plus 12).

2. Tap the intersections: $(-2, 4)$ and $(6, 36)$. The number is positive.

3. The number is 6. Check: Type `6^2`: it returns 36 and type `4*6+12`: it returns 36.

Answer: 6

55. Medium · Desmos

$$xy = 48$$
$$3x - y = 0$$

If (x, y) is a solution to the system above and $x > 0$, what is the value of y?

1. From $3x - y = 0$ we get $y = 3x$. Type `x*y=48` and `y=3x`.

2. Tap the two intersections: $(-4, -12)$ and $(4, 12)$. We need $x > 0$.

3. So $y = 12$. Check: Type `4*12`: it returns 48.

Answer: 12

56. Medium · Desmos

If $2^{x+3} - 2^x = 56$, what is the value of x?

1. Type `y=2^(x+3)-2^x` and `y=56`.

2. Tap the intersection: $(3, 56)$. So $x = 3$.

3. Check: Type `2^(3+3)-2^3`: it returns 56.

Answer: 3

57. Medium · Desmos

What is the product of the positive solutions to the equation $x^4 - 13x^2 + 36 = 0$?

1. Type `y=x^4-13x^2+36`.

2. Tap the four zeros: $(-3, 0)$, $(-2, 0)$, $(2, 0)$ and $(3, 0)$.

3. The positive solutions are 2 and 3: Type `2*3`: it returns 6.

Answer: 6